

DBpedia-based Movie Ontology: Evidence-driven Design

This design is grounded in the existing crawled dataset. Domain facts are not invented. The native build, application and query modes share these version-three definitions and canonical exports.

1. Findings and semantic decisions

- Reuse `dbo:Film`, `Person`, `Actor`, `Writer`, `MovieDirector`, `ScreenWriter`, `Producer`, `Company`, `Organisation`, `Award`, `MovieGenre`, `Genre`, `Country`, `Language`, `Work`, `Artist` and `Agent`. `Actor` follows the original `dbo:starring` range; no equivalent-class override of `dbo:Actor`.
- Reuse `dbo:director`, `starring`, `writer`, `producer`, `productionCompany`, `award`, `genre`, `country`, `language` and `runtime`. Preserve reference domain/range: `starring` ranges over `Actor`, `producer` over `Agent`, `productionCompany` over `Company`.
- `dbo:MovieDirector`, `ScreenWriter` and `Producer` are reused for occupation-level inferences from the corresponding credits. One-way subclass axioms suffice; their global definitions are not overridden. – `dbo:runtime` is seconds (`xsd:double`); *Inception* has 8880 seconds, converted from the crawled 148 minutes. Titles reuse `rdfs:label` (annotation property, not a custom datatype property).
- *P577* has varying precision. Keep a documented local `releaseYear` summary instead of inventing a full `dbo:releaseDate`. Budget, gross, IMDb, distributors and ratings are excluded because they are not in the collected field set.
- Remove unsupported Documentary/Fiction hierarchy and FeatureFilm assertions: Wikidata Q11424 means film and does not justify feature-film classification. Award labels alone do not justify treating every unrecognized award as a film award.
- `ActionGenre` and `DramaGenre` are transparent label-normalization buckets, not source-provided OWL classes. The author-specified classification rule is separate from the crawled facts; fiction/non-fiction is not guessed.
- Contribution and four role individuals represent *P57/P161/P58/P162* credits. Each record has exactly one person, film and role. Missing values remain an open-world validation issue, not automatic inconsistency.
- `MultiCreditContributor` / `ThreeCreditContributor` mean at least two/three distinct credit records. Different roles provide witnesses; these class names do not prove exactly two/three unique roles across an entire career. More precise display names: `MultiCreditContributor` / `ThreeCreditContributor`.
- `MultiGenreFilm`, `MultiAwardFilm` and `FrequentProductionCompany` are not included as inferred classes: different genre/award/film IRIs alone do not establish inequality. `COUNT DISTINCT` can answer a database-style question, but cannot be presented as OWL DL entailment.

Reference scope

The official 2016–10 reference and official extraction-framework development vocabulary (790 classes) were checked. Semantic alternatives reviewed include `Actor`, `MovieDirector`, `ScreenWriter`, `Producer`, `Writer`, `ArtisticGenre`, `LiteraryGenre`, `MovieGenre`; `dbo:occupation` ranges over `PersonFunction`, while `dbo:role` and `artistFunction` are strings and cannot replace the local credit-to-role entity link. `dbo:year` (`gYear`) is generic, whereas `ex:releaseYear` explicitly represents an integer earliest-release summary; it does not claim equivalence to `dbo:year` or a complete `releaseDate`. The archived reference hash is: `a0ca30c8f25cf2d2957639dcad4fd1115f37a856d6d542fd50bba3bf9fa6bf9f`. The current official runtime, `starring` and `MovieGenre` pages were cross-checked. Absence in this version is not proof that no equivalent vocabulary exists anywhere; local `CREATE/EXTEND` decisions are scoped to the checked DBpedia vocabulary and the stated meanings.

Target size

The overview uses 14 main concepts: `dbo:Film`, `Person`, `Actor`, `Company`, `Award`, `MovieGenre`, `Country`, `Language`; `ex:Contribution`, `ContributionRole`, `Filmmaker`, `ActionFilm`, `AwardWinningFilm` and `MultiCreditContributor`. Supporting hierarchy, provenance and inferred subsets are listed separately. There are 10 domain inferred classes, including seven additional classes requested in section 13, plus four supporting credit subclasses and the reused `Actor` inference. This deliberately exceeds the approximate 5–8 final target to satisfy the stronger requirement of seven additional supported classes; no empty classes were added to reach a count.

2. TABLE 1 – Class inventory

Class / URI	DBpedia equivalent	Action	Parent	Explicit / Inferred	Manchester definition	Source and meaning
<code>dbo:Work</code>	<code>dbo:Work</code>	REUSE	–	Base / hierarchy	–	DBpedia supertype of supported entities; Reuse DBpedia meaning and its supported superclass; no local clone.
<code>dbo:Film</code>	<code>dbo:Film</code>	REUSE	<code>dbo:Work</code>	Base / hierarchy	–	P31; Reuse DBpedia meaning and its supported superclass; no local clone.
<code>dbo:Agent</code>	<code>dbo:Agent</code>	REUSE	–	Base / hierarchy	–	DBpedia supertype of supported entities; Reuse DBpedia meaning and its supported superclass; no local clone.

Class / URI	DBpedia equivalent	Action	Parent	Explicit / Inferred	Manchester definition	Source and meaning
dbo:Person	dbo:Person	REUSE	dbo:Agent	Base / hierarchy	—	P31=Q5; P57/P161/P58/P162; Reuse DBpedia meaning and its supported superclass; no local clone.
dbo:Artist	dbo:Artist	REUSE	dbo:Person	Base / hierarchy	—	DBpedia supertype of supported entities; Reuse DBpedia meaning and its supported superclass; no local clone.
dbo:Actor	dbo:Actor	REUSE	dbo:Artist	Inferred	—	P161; Reuse DBpedia meaning and its supported superclass; no local clone.
dbo:Writer	dbo:Writer	REUSE	dbo:Person	Inferred	—	P58 via ScreenWriter; Reuse DBpedia meaning and its supported superclass; no local clone.
dbo:Movie Director	dbo:Movie Director	REUSE	dbo:Person	Inferred	—	P57; Reuse DBpedia meaning and its supported superclass; no local clone.

Class / URI	DBpedia equivalent	Action	Parent	Explicit / Inferred	Manchester definition	Source and meaning
dbo:Screen Writer	dbo:Screen Writer	REUSE	dbo:Writer	Inferred	—	P58; Reuse DBpedia meaning and its supported superclass; no local clone.
dbo:Producer	dbo:Producer	REUSE	dbo:Person	Inferred	—	P162; Reuse DBpedia meaning and its supported superclass; no local clone.
dbo:Organisation	dbo:Organisation	REUSE	dbo:Agent	Base / hierarchy	—	DBpedia supertype of supported entities; Reuse DBpedia meaning and its supported superclass; no local clone.
dbo:Company	dbo:Company	REUSE	dbo:Organisation	Base / hierarchy	—	P272; Reuse DBpedia meaning and its supported superclass; no local clone.
dbo:Genre	dbo:Genre	REUSE	—	Base / hierarchy	—	P136; Reuse DBpedia meaning and its supported superclass; no local clone.
dbo:Movie Genre	dbo:Movie Genre	REUSE	dbo:Genre	Base / hierarchy	—	P136 in film context; Reuse DBpedia meaning and its supported superclass; no local clone.

Class / URI	DBpedia equivalent	Action	Parent	Explicit / Inferred	Manchester definition	Source and meaning
dbo:Award	dbo:Award	REUSE	—	Base / hierarchy	—	P166; Reuse DBpedia meaning and its supported superclass; no local clone.
dbo:Country	dbo:Country	REUSE	—	Base / hierarchy	—	P495; Reuse DBpedia meaning and its supported superclass; no local clone.
dbo:Language	dbo:Language	REUSE	—	Base / hierarchy	—	P364; Reuse DBpedia meaning and its supported superclass; no local clone.
ex:Contribution	No matching meaning in checked reference	CREATE	—	Base / hierarchy	—	P57/P161/P58/P162; A local credit association with one person, film and role; not the person or film itself.
ex:ContributionRole	No matching meaning in checked reference	CREATE	—	Base / hierarchy	—	The four crawled credit predicates; Local controlled role values, not subclasses of Person.
ex:SourceSnapshot	No matching meaning in checked reference	CREATE	—	Base / hierarchy	—	snapshots. json URL/time/hash; A retrieved response record, not a domain entity.

Class / URI	DBpedia equivalent	Action	Parent	Explicit / Inferred	Manchester definition	Source and meaning
void:Dataset	void:Dataset	REUSE	—	Base / hierarchy	—	Published collection of crawled film records; Reuse VOID dataset, remove ex:Dataset clone.
ex:Action Genre	No matching meaning in checked reference	EXTEND	dbo:Movie Genre	Base / hierarchy	—	P136 target English label; Local normalized film-genre bucket selected by matching the crawled English label; no DBpedia class of this narrow meaning in reference module.
ex:Drama Genre	No matching meaning in checked reference	EXTEND	dbo:Movie Genre	Base / hierarchy	—	P136 target English label; Local normalized film-genre bucket selected by matching the crawled English label; no DBpedia class of this narrow meaning in reference module.
ex:Acting Contribution	No matching meaning in checked reference	EXTEND	ex:Contribution	Inferred	(ex:Contribution and ex:hasRole value ex:ActorRole)	Credit claim corresponding to Actor; Local credit record classified by a fixed role; DBpedia Actor describes a person, not this record.

Class / URI	DBpedia equivalent	Action	Parent	Explicit / Inferred	Manchester definition	Source and meaning
ex:DirectingContribution	No matching meaning in checked reference	EXTEND	ex:Contribution	Inferred	(ex:Contribution and ex:hasRole value ex:DirectorRole)	Credit claim corresponding to Director; Local credit record classified by a fixed role; DBpedia Actor describes a person, not this record.
ex:WritingContribution	No matching meaning in checked reference	EXTEND	ex:Contribution	Inferred	(ex:Contribution and ex:hasRole value ex:WriterRole)	Credit claim corresponding to Writer; Local credit record classified by a fixed role; DBpedia Actor describes a person, not this record.
ex:ProducingContribution	No matching meaning in checked reference	EXTEND	ex:Contribution	Inferred	(ex:Contribution and ex:hasRole value ex:ProducerRole)	Credit claim corresponding to Producer; Local credit record classified by a fixed role; DBpedia Actor describes a person, not this record.

Class / URI	DBpedia equivalent	Action	Parent	Explicit / Inferred	Manchester definition	Source and meaning
ex:Filmmaker	No matching meaning in checked reference	EXTEND	dbo:Person	Inferred	(dbo:Person and (ex:hasContribution some ex:DirectingContribution or ex:hasContribution some ex:WritingContribution or ex:hasContribution some ex:ProducingContribution))	P57/P161/P58/P162; P166 when award-based; Local dataset-specific OWL-defined subset; not a replacement for a DBpedia base class.
ex:Action Film	No matching meaning in checked reference	EXTEND	dbo:Film	Inferred	(dbo:Film and dbo:genre some ex:ActionGenre)	P136/P166; Local dataset-specific OWL-defined subset; not a replacement for a DBpedia base class.
ex:Award WinningFilm	No matching meaning in checked reference	EXTEND	dbo:Film	Inferred	(dbo:Film and dbo:award some dbo:Award)	P136/P166; Local dataset-specific OWL-defined subset; not a replacement for a DBpedia base class.

Class / URI	DBpedia equivalent	Action	Parent	Explicit / Inferred	Manchester definition	Source and meaning
ex:MultiCreditContributor	No matching meaning in checked reference	EXTEND	dbo:Person	Inferred	(dbo:Person and ex:hasContribution min 2 ex:Contribution)	P57/P161/P58/P162; P166 when award-based; At least two semantically distinct credit records. Different roles prove their distinctness; the name is shorthand and does not exclude repeated same-role credits.
ex:ThreeCreditContributor	No matching meaning in checked reference	EXTEND	dbo:Person	Inferred	(dbo:Person and ex:hasContribution min 3 ex:Contribution)	P57/P161/P58/P162; P166 when award-based; At least three distinct credit records; three different role fillers prove distinctness. Not exactly three roles or an exhaustive career description.

Class / URI	DBpedia equivalent	Action	Parent	Explicit / Inferred	Manchester definition	Source and meaning
ex:Writer Director	No matching meaning in checked reference	EXTEND	dbo:Person	Inferred	(dbo:Movie Director and dbo:Screen Writer)	P57/P161/P58/P162; P166 when award-based; A person with both directing and writing credits in this dataset; classification is never manually asserted.
ex:Actor Filmmaker	No matching meaning in checked reference	EXTEND	dbo:Person	Inferred	(dbo:Actor and ex:Filmmaker)	P57/P161/P58/P162; P166 when award-based; Local dataset-specific OWL-defined subset; not a replacement for a DBpedia base class.
ex:Award WinningFilmmaker	No matching meaning in checked reference	EXTEND	dbo:Person	Inferred	(ex:Filmmaker and dbo:award some dbo:Award)	P57/P161/P58/P162; P166 when award-based; Local dataset-specific OWL-defined subset; not a replacement for a DBpedia base class.

Class / URI	DBpedia equivalent	Action	Parent	Explicit / Inferred	Manchester definition	Source and meaning
ex:Award WinningActionFilm	No matching meaning in checked reference	EXTEND	dbo:Film	Inferred	(ex:Action Film and ex:Award WinningFilm)	P136/P166; Local dataset-specific OWL-defined subset; not a replacement for a DBpedia base class.
ex:Genre Crossing Film	No matching meaning in checked reference	EXTEND	dbo:Film	Inferred	(dbo:Film and dbo:genre some ex:Action Genre and dbo:genre some ex:Drama Genre)	P136/P166; Film carrying both action and drama genre memberships; this is not an unproved two-distinct-genre cardinality claim.

3. TABLE 2 – Object properties

Property / URI	Reuse / Create	Domain	Range	Inverse	Subproperty	Characteristics	Source / meaning
dbo:director	REUSE	dbo:Film	dbo:Person	–	–	No extra characteristic	P57
dbo:starring	REUSE	dbo:Work	dbo:Actor	–	–	No extra characteristic	P161
dbo:writer	REUSE	dbo:Work	dbo:Person	–	–	No extra characteristic	P58
dbo:producer	REUSE	dbo:Work	dbo:Agent	–	–	No extra characteristic	P162
dbo:productionCompany	REUSE	dbo:Work	dbo:Company	–	–	No extra characteristic	P272
dbo:award	REUSE	–	dbo:Award	–	–	No extra characteristic	P166
dbo:genre	REUSE	–	dbo:Genre	–	–	No extra characteristic	P136

Property / URI	Reuse / Create	Domain	Range	Inverse	Subproperty	Characteristics	Source / meaning
dbo:country	REUSE	—	dbo:Country	—	—	No extra characteristic	P495
dbo:language	REUSE	—	dbo:Language	—	—	No extra characteristic	P364
ex:hasContribution	CREATE	dbo:Person	ex:Contribution	ex:contributionBy	—	No extra characteristic	Derived credit association
ex:contributionBy	CREATE	ex:Contribution	dbo:Person	—	—	Functional	P57/P161/ P58/P162
ex:contributionTo	CREATE	ex:Contribution	dbo:Film	ex:contributionOf	—	Functional	Containing source film
ex:contributionOf	CREATE	dbo:Film	ex:Contribution	—	—	No extra characteristic	Inverse of contributionTo
ex:hasRole	CREATE	ex:Contribution	ex:ContributionRole	—	—	Functional	Credit predicate to controlled role
ex:contributedTo	CREATE	dbo:Person	dbo:Film	—	—	No extra characteristic	OWL property chain
ex:directed	CREATE	dbo:Person	dbo:Film	dbo:director	ex:contributedTo	No extra characteristic	Inverse P57
ex:actedIn	CREATE	dbo:Actor	dbo:Film	dbo:starring	ex:contributedTo	No extra characteristic	Inverse P161
ex:productionOf	CREATE	dbo:Company	dbo:Work	dbo:productionCompany	—	No extra characteristic	Inverse P272
ex:sourceSnapshot	CREATE	—	ex:SourceSnapshot	—	—	No extra characteristic	Crawled entity response provenance

No symmetric, transitive or inverse-functional flags are added. contributedTo is non-simple because it has a property chain; cardinality is placed on the simple hasContribution property, not contributedTo. Standard owl:sameAs and prov:wasDerivedFrom remain as identity/provenance predicates; they are not custom domain properties.

4. TABLE 3 – Data properties

Property	Reuse / Create	Domain	Datatype	Cardinality	Crawled source	Meaning
dbo:runtime	REUSE	dbo:Work	xsd:double	No global max; one normalized value in this export	P2047; normalize to seconds	Seconds
ex:release Year	CREATE	dbo:Film	xsd:integer	No global max; one normalized value in this export	Earliest Gregorian P577 year, precision >=9	Earliest Gregorian P577 year, precision >=9
ex:sourceUrl	CREATE	ex:Source Snapshot	xsd:anyURI	No global max; one normalized value in this export	Snapshot manifest sourceUrl	Snapshot manifest sourceUrl
ex:retrieved At	CREATE	ex:Source Snapshot	xsd:dateTime	No global max; one normalized value in this export	Snapshot manifest retrievedAt	Snapshot manifest retrievedAt
ex:sha256	CREATE	ex:Source Snapshot	xsd:string	No global max; one normalized value in this export	Snapshot manifest sha256	Snapshot manifest sha256

rdfs:label is reused as an annotation property with language-tagged text; it is intentionally not declared owl:DatatypeProperty. There are 19 object + 5 datatype properties in the design, excluding built-in and metadata predicates.

5. TABLE 4 – Inferred classes and actual results

Class	DBpedia equivalent	OWL restriction	Real example	Initial facts	New knowledge / local count
dbo:Actor	dbo:Actor	Range(dbo:starrier) = Starrier	Boyd (res:person-Q1016897)	Source triples shown below	0 asserted → 769 newly entailed
dbo:Writer	dbo:Writer	Credit existential / reused superclass inference	Philippa Boyens (res:person-Q116854)	Source triples shown below	0 asserted → 34 newly entailed
dbo:Movie Director	dbo:Movie Director	Credit existential / reused superclass inference	Damien Chazelle (res:person-Q18350026)	Source triples shown below	0 asserted → 17 newly entailed
dbo:Screen Writer	dbo:Screen Writer	Credit existential / reused superclass inference	Philippa Boyens (res:person-Q116854)	Source triples shown below	0 asserted → 34 newly entailed

Class	DBpedia equivalent	OWL restriction	Real example	Initial facts	New knowledge / local count
dbo:Producer	dbo:Producer	Credit existential / reused superclass inference	Fran Walsh (res:person-Q116861)	Source triples shown below	0 asserted → 57 newly entailed
ex:Acting Contribution	No equivalent checked	(ex:Contribution and ex:hasRole value ex:ActorRole)	Alien – actor – Sigourney Weaver (res:contribution-Q103569-Q102124-actor)	Source triples shown below	0 asserted → 855 newly entailed
ex:Directing Contribution	No equivalent checked	(ex:Contribution and ex:hasRole value ex:DirectorRole)	Alien – director – Ridley Scott (res:contribution-Q103569-Q56005-director)	Source triples shown below	0 asserted → 31 newly entailed
ex:Writing Contribution	No equivalent checked	(ex:Contribution and ex:hasRole value ex:WriterRole)	Alien – writer – Ronald Shusett (res:contribution-Q103569-Q3056094-writer)	Source triples shown below	0 asserted → 51 newly entailed
ex:Producing Contribution	No equivalent checked	(ex:Contribution and ex:hasRole value ex:ProducerRole)	Alien – producer – Gordon Carroll (res:contribution-Q103569-Q1537983-producer)	Source triples shown below	0 asserted → 73 newly entailed
ex:Filmmaker	No equivalent checked	(dbo:Person and (ex:hasContribution some ex:DirectingContribution or ex:hasContribution some ex:WritingContribution or ex:hasContribution some ex:ProducingContribution))	Philippa Boyens (res:person-Q116854)	Source triples shown below	0 asserted → 89 newly entailed
ex:ActionFilm	No equivalent checked	(dbo:Film and dbo:genre some ex:ActionGenre)	Alien (res:film-Q103569)	Source triples shown below	0 asserted → 12 newly entailed

Class	DBpedia equivalent	OWL restriction	Real example	Initial facts	New knowledge / local count
ex:Award WinningFilm	No equivalent checked	(dbo:Film and dbo:award some dbo:Award)	Alien (res:film-Q103569)	Source triples shown below	0 asserted → 26 newly entailed
ex:MultiCredit Contributor	No equivalent checked	(dbo:Person and ex:hasContribution min 2 ex:Contribution)	Fran Walsh (res:person-Q116861)	Source triples shown below	0 asserted → 17 newly entailed
ex:Three CreditContributor	No equivalent checked	(dbo:Person and ex:hasContribution min 3 ex:Contribution)	Christopher Nolan (res:person-Q25191)	Source triples shown below	0 asserted → 7 newly entailed
ex:Writer Director	No equivalent checked	(dbo:Movie Director and dbo:Screen Writer)	Damien Chazelle (res:person-Q18350026)	Source triples shown below	0 asserted → 10 newly entailed
ex:Actor Filmmaker	No equivalent checked	(dbo:Actor and ex:Filmmaker)	Quentin Tarantino (res:person-Q3772)	Source triples shown below	0 asserted → 7 newly entailed
ex:Award WinningFilmmaker	No equivalent checked	(ex:Filmmaker and dbo:award some dbo:Award)	Philippa Boyens (res:person-Q116854)	Source triples shown below	0 asserted → 65 newly entailed
ex:Award WinningAction Film	No equivalent checked	(ex:ActionFilm and ex:Award WinningFilm)	Alien (res:film-Q103569)	Source triples shown below	0 asserted → 10 newly entailed
ex:Genre CrossingFilm	No equivalent checked	(dbo:Film and dbo:genre some ex:ActionGenre and dbo:genre some ex:DramaGenre)	Pulp Fiction (res:film-Q104123)	Source triples shown below	0 asserted → 8 newly entailed

6. TABLE 5 – Multi-step chains

Initial facts	Step 1	Step 2	Step 3	Final inference
Film starring Person	dbo:starring range → Actor	Actor Artist	Artist Person	Person Agent
Contribution hasRole DirectorRole	hasValue → DirectingContribution	inverse contributionBy → hasContribution	some DirectingContribution → Filmmaker	Person/Agent hierarchy
Film genre typed ActionGenre	ActionGenre MovieGenre	MovieGenre Genre	some ActionGenre → ActionFilm	Parent genre query matches
Person directing + writing credits	Role → Directing/WritingContribution	Inverse produces hasContribution	Two existential restrictions → WriterDirector	Filmmaker also follows

Initial facts	Step 1	Step 2	Step 3	Final inference
Person with three credit roles	Different controlled roles	Functional hasRole proves records different	min 3 $\rightarrow$ ThreeCreditContributor	min 2 $\rightarrow$ MultiCreditContributor
Film ActionGenre + award	Genre hierarchy and award range	ActionFilm / AwardWinningFilm	Intersection $\rightarrow$ AwardWinningActionFilm	New type absent before
Credit person and film links	Inverse hasContribution	hasContribution contributionTo $\rightarrow$ contributedTo	dbo:director inverse $\rightarrow$ directed	directed contributedTo

7. TABLE 6 – Executed SPARQL competency questions

# / Question	SPARQL file	Direct / Reasoned	Before $\rightarrow$ After rows	Reasoning value
1. List films	queries/01.rq	A	30 $\rightarrow$ 30	Source facts
2. Inception: year, duration and director	queries/02.rq	A	1 $\rightarrow$ 1	Source facts
3. Directors of Inception	queries/03.rq	A	1 $\rightarrow$ 1	Source facts
4. Inception credits and roles	queries/04.rq	A	25 $\rightarrow$ 25	Source facts
5. Films directed by Nolan	queries/05.rq	A	8 $\rightarrow$ 8	Source facts
6. Companies credited on Inception	queries/06.rq	A	4 $\rightarrow$ 4	Source facts
7. Awards of Inception	queries/07.rq	A	7 $\rightarrow$ 7	Source facts
8. Runtime in seconds	queries/08.rq	A	1 $\rightarrow$ 1	Source facts
9. Film genres through the parent class	queries/09.rq	B	0 $\rightarrow$ 75	Hierarchy closure
10. Persons through Agent hierarchy	queries/10.rq	B	0 $\rightarrow$ 851	Hierarchy closure
11. Production companies	queries/11.rq	B	45 $\rightarrow$ 45	Hierarchy closure
12. Film subclasses	queries/12.rq	B	0 $\rightarrow$ 4	Hierarchy closure
13. Actors through Artist hierarchy	queries/13.rq	B	0 $\rightarrow$ 769	Hierarchy closure
14. Inferred Actors	queries/14.rq	C	0 $\rightarrow$ 769	Shared inferred types / relationships
15. Inferred Filmmakers	queries/15.rq	C	0 $\rightarrow$ 89	Shared inferred types / relationships

# / Question	SPARQL file	Direct / Reasoned	Before → After rows	Reasoning value
16. Inferred Action films	queries/16.rq	C	0 → 12	Shared inferred types / relation ships
17. Award-winning films	queries/17.rq	C	0 → 26	Shared inferred types / relation ships
18. People with at least two provably distinct credits	queries/18.rq	C	0 → 17	Shared inferred types / relation ships
19. People with at least three provably distinct credits	queries/19.rq	C	0 → 7	Shared inferred types / relation ships
20. Writer-directors	queries/20.rq	C	0 → 10	Shared inferred types / relation ships
21. Actor-filmmakers	queries/21.rq	C	0 → 7	Shared inferred types / relation ships
22. Award-winning filmmakers	queries/22.rq	C	0 → 65	Shared inferred types / relation ships
23. Award-winning action films	queries/23.rq	C	0 → 10	Shared inferred types / relation ships
24. Action/drama crossing films	queries/24.rq	C	0 → 8	Shared inferred types / relation ships
25. Nolan contributions through property chain	queries/25.rq	C	0 → 8	Shared inferred types / relation ships
26. Nolan films through inverse director	queries/26.rq	C	0 → 8	Shared inferred types / relation ships
27. Types present only after reasoning	queries/27.rq	C	0 → 1261	Shared inferred types / relation ships

Group A queries use asserted.ttl. Group B needs schema and hierarchy closure; group C uses inferred.ttl together with schema + asserted data. Query 27 uses the named graphs in before_after.trig. Entity-result queries restrict results to local resource IRIs and exclude sameAs aliases; the class-count table uses the same local identity scope. Query 8 returns a literal and query 12 returns class URIs, so they use the corresponding value/class scope.

8. TABLE 7 – Five strongest verified demos

Question	Initial data	DBpedia reuse	Custom rule	Inferred result	Why it matters
Who is an Actor?	Real source triples below	dbo:Film / Person / Actor / genre / award	starring range + hierarchy	769 newly entailed local members	Reusable explicit semantics; before/after verified
Who is a Filmmaker?	Real source triples below	dbo:Film / Person / Actor / genre / award	(dbo:Person and (ex:hasContribution some ex:DirectingContribution or ex:hasContribution some ex:WritingContribution or ex:hasContribution some ex:ProducingContribution))	89 newly entailed local members	Reusable explicit semantics; before/after verified
Which films are ActionFilm?	Real source triples below	dbo:Film / Person / Actor / genre / award	(dbo:Film and dbo:genre some ex:ActionGenre)	12 newly entailed local members	Reusable explicit semantics; before/after verified
Who has three provably distinct credits?	Real source triples below	dbo:Film / Person / Actor / genre / award	(dbo:Person and ex:hasContribution min 3 ex:Contribution)	7 newly entailed local members	Reusable explicit semantics; before/after verified
Which action films received an award?	Real source triples below	dbo:Film / Person / Actor / genre / award	(ex:ActionFilm and ex:Award WinningFilm)	10 newly entailed local members	Reusable explicit semantics; before/after verified

MultiGenreFilm was a preferred demo, but the real DL run inferred zero members. The fourth demo uses genuine min 3 reasoning instead. This meets the cardinality objective without inventing genre inequality.

9. Five complete Semantic Web demonstrations

Who is an Actor?

QUESTION: Who is an Actor?

INITIAL TRIPLES:

```

res:contribution-Q104123-Q1016897-actor ex:contributionBy res:person-Q1016897 .
res:contribution-Q104123-Q1016897-actor ex:contributionTo res:film-Q104123 .
res:contribution-Q104123-Q1016897-actor ex:hasRole ex:ActorRole .
res:contribution-Q104123-Q1016897-actor
  ⇨ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:film-Q104123 dbo:starring res:person-Q1016897 .
res:person-Q1016897 ex:hasContribution res:contribution-Q104123-Q1016897-actor .
res:person-Q1016897 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:Person .

```

DBPEDIA VOCABULARY: dbo:Film, dbo:Person, dbo:Actor and the source-supported properties in Table 2.

ONTOLOGY RULE: `dbo:starring rdfs:range dbo:Actor`; Actor subclass Artist subclass Person subclass Agent.

REASONING: The source relation supplies existential witnesses. Supporting credit classes follow role hasValue restrictions. For min 3, the three role individuals are distinct by their controlled meanings; functional hasRole forces the corresponding credits to be distinct. Hermit classified the complete graph.

INFERRED KNOWLEDGE: `res:person-Q1016897 rdf:type dbo:Actor`. This type has zero direct assertions in the input graph.

SPARQL:

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a dbo:Actor . FILTER(STRSTARTS(STR(?s),
  ↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/") ) OPTIONAL {
  ↪ ?s rdfs:label ?label } } ORDER BY ?s
```

RESULT: 769 newly entailed local individuals; example Burr Steers (`res:person-Q1016897`).

WHY THIS DEMONSTRATES SEMANTIC WEB: The ontology makes the classification and shared vocabulary explicit and reusable across consumers. SQL joins, recursive queries and rule systems can reproduce many of these answers; the advantage is declared semantics, interoperable identifiers, open-world entailment and consistency checking, not an inability of databases to compute them.

Who is a Filmmaker?

QUESTION: Who is a Filmmaker?

INITIAL TRIPLES:

```
res:contribution-Q13417189-Q25191-director ex:contributionBy res:person-Q25191 .
res:contribution-Q13417189-Q25191-director ex:contributionTo res:film-Q13417189 .
res:contribution-Q13417189-Q25191-director ex:hasRole ex:DirectorRole .
res:contribution-Q13417189-Q25191-director
  ↪ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:contribution-Q13417189-Q25191-producer ex:contributionBy res:person-Q25191 .
res:contribution-Q13417189-Q25191-producer ex:contributionTo res:film-Q13417189 .
res:contribution-Q13417189-Q25191-producer ex:hasRole ex:ProducerRole .
res:contribution-Q13417189-Q25191-producer
  ↪ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:contribution-Q13417189-Q25191-writer ex:contributionBy res:person-Q25191 .
res:contribution-Q13417189-Q25191-writer ex:contributionTo res:film-Q13417189 .
res:contribution-Q13417189-Q25191-writer ex:hasRole ex:WriterRole .
res:contribution-Q13417189-Q25191-writer
  ↪ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:person-Q25191 dbo:award res:award-Q102427 .
res:person-Q25191 dbo:award res:award-Q103360 .
res:person-Q25191 dbo:award res:award-Q105447 .
res:person-Q25191 dbo:award res:award-Q1056240 .
res:person-Q25191 dbo:award res:award-Q12143912 .
res:person-Q25191 dbo:award res:award-Q12201477 .
res:person-Q25191 dbo:award res:award-Q16058279 .
res:person-Q25191 dbo:award res:award-Q3405197 .
res:person-Q25191 dbo:award res:award-Q586356 .
res:person-Q25191 dbo:award res:award-Q787131 .
res:person-Q25191 dbo:award res:award-Q833154 .
res:person-Q25191 dbo:award res:award-Q833163 .
res:person-Q25191 ex:hasContribution res:contribution-Q13417189-Q25191-director .
```

```
res:person-Q25191 ex:hasContribution res:contribution-Q13417189-Q25191-producer .
res:person-Q25191 ex:hasContribution res:contribution-Q13417189-Q25191-writer .
res:person-Q25191 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:Person .
```

DBPEDIA VOCABULARY: dbo:Film, dbo:Person, dbo:Actor and the source-supported properties in Table 2.

ONTOLOGY RULE: (dbo:Person and (ex:hasContribution some ex:DirectingContribution or ex:hasContribution some ex:WritingContribution or ex:hasContribution some ex:ProducingContribution))

REASONING: The source relation supplies existential witnesses. Supporting credit classes follow role hasValue restrictions. For min 3, the three role individuals are distinct by their controlled meanings; functional hasRole forces the corresponding credits to be distinct. Hermit classified the complete graph.

INFERRED KNOWLEDGE: res:person-Q25191 rdf:type ex:Filmmaker. This type has zero direct assertions in the input graph.

SPARQL:

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a ex:Filmmaker . FILTER(STRSTARTS(STR(?s),
  ↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
  ↪ ?s rdfs:label ?label } } ORDER BY ?s
```

RESULT: 89 newly entailed local individuals; example Christopher Nolan (res:person-Q25191).

WHY THIS DEMONSTRATES SEMANTIC WEB: The ontology makes the classification and shared vocabulary explicit and reusable across consumers. SQL joins, recursive queries and rule systems can reproduce many of these answers; the advantage is declared semantics, interoperable identifiers, open-world entailment and consistency checking, not an inability of databases to compute them.

Which films are ActionFilm?

QUESTION: Which films are ActionFilm?

INITIAL TRIPLES:

```
res:film-Q103569 dbo:award res:award-Q1131772 .
res:film-Q103569 dbo:award res:award-Q1257399 .
res:film-Q103569 dbo:award res:award-Q1265702 .
res:film-Q103569 dbo:award res:award-Q3414212 .
res:film-Q103569 dbo:award res:award-Q393686 .
res:film-Q103569 dbo:award res:award-Q739633 .
res:film-Q103569 dbo:genre res:genre-Q102260466 .
res:film-Q103569 dbo:genre res:genre-Q10663882 .
res:film-Q103569 dbo:genre res:genre-Q11304653 .
res:film-Q103569 dbo:genre res:genre-Q1200678 .
res:film-Q103569 dbo:genre res:genre-Q1342372 .
res:film-Q103569 dbo:genre res:genre-Q157394 .
res:film-Q103569 dbo:genre res:genre-Q1776156 .
res:film-Q103569 dbo:genre res:genre-Q188473 .
res:film-Q103569 dbo:genre res:genre-Q200092 .
res:film-Q103569 dbo:genre res:genre-Q20656232 .
res:film-Q103569 dbo:genre res:genre-Q2484376 .
res:film-Q103569 dbo:genre res:genre-Q319221 .
res:film-Q103569 dbo:genre res:genre-Q471839 .
res:film-Q103569 dbo:starring res:person-Q102124 .
```

```

res:film-Q103569 dbo:starring res:person-Q200405 .
res:film-Q103569 dbo:starring res:person-Q223091 .
res:film-Q103569 dbo:starring res:person-Q266270 .
res:film-Q103569 dbo:starring res:person-Q283872 .
res:film-Q103569 dbo:starring res:person-Q3047302 .
res:film-Q103569 dbo:starring res:person-Q314290 .
res:film-Q103569 dbo:starring res:person-Q320093 .
res:film-Q103569 dbo:starring res:person-Q323185 .
res:film-Q103569 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:Film .
res:genre-Q102260466 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q10663882 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q11304653 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q1200678 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q1342372 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q157394 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q1776156 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q188473 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q188473 http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:ActionGenre .
res:genre-Q200092 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q20656232 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q20656232 http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:ActionGenre
↪ .
res:genre-Q2484376 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q319221 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q471839 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .

```

DBPEDIA VOCABULARY: dbo:Film, dbo:Person, dbo:Actor and the source-supported properties in Table 2.

ONTOLOGY RULE: (dbo:Film and dbo:genre some ex:ActionGenre)

REASONING: The source relation supplies existential witnesses. Supporting credit classes follow role hasValue restrictions. For min 3, the three role individuals are distinct by their controlled meanings; functional hasRole forces the corresponding credits to be distinct. Hermit classified the complete graph.

INFERRED KNOWLEDGE: res:film-Q103569 rdf:type ex:ActionFilm. This type has zero direct assertions in the input graph.

SPARQL:

```

PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a ex:ActionFilm . FILTER(STRSTARTS(STR(?s),
↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
↪ ?s rdfs:label ?label } } ORDER BY ?s

```

RESULT: 12 newly entailed local individuals; example Alien (res:film-Q103569).

WHY THIS DEMONSTRATES SEMANTIC WEB: The ontology makes the classification and shared vocabulary explicit and reusable across consumers. SQL joins, recursive queries and rule systems can reproduce many of these answers; the advantage is declared semantics, interoperable identifiers, open-world entailment and consistency checking, not an inability of databases to compute them.

Who has three provably distinct credits?

QUESTION: Who has three provably distinct credits?

INITIAL TRIPLES:

```
res:contribution-Q13417189-Q25191-director ex:contributionBy res:person-Q25191 .
res:contribution-Q13417189-Q25191-director ex:contributionTo res:film-Q13417189 .
res:contribution-Q13417189-Q25191-director ex:hasRole ex:DirectorRole .
res:contribution-Q13417189-Q25191-director
  ↪ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:contribution-Q13417189-Q25191-producer ex:contributionBy res:person-Q25191 .
res:contribution-Q13417189-Q25191-producer ex:contributionTo res:film-Q13417189 .
res:contribution-Q13417189-Q25191-producer ex:hasRole ex:ProducerRole .
res:contribution-Q13417189-Q25191-producer
  ↪ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:contribution-Q13417189-Q25191-writer ex:contributionBy res:person-Q25191 .
res:contribution-Q13417189-Q25191-writer ex:contributionTo res:film-Q13417189 .
res:contribution-Q13417189-Q25191-writer ex:hasRole ex:WriterRole .
res:contribution-Q13417189-Q25191-writer
  ↪ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:person-Q25191 dbo:award res:award-Q102427 .
res:person-Q25191 dbo:award res:award-Q103360 .
res:person-Q25191 dbo:award res:award-Q105447 .
res:person-Q25191 dbo:award res:award-Q1056240 .
res:person-Q25191 dbo:award res:award-Q12143912 .
res:person-Q25191 dbo:award res:award-Q12201477 .
res:person-Q25191 dbo:award res:award-Q16058279 .
res:person-Q25191 dbo:award res:award-Q3405197 .
res:person-Q25191 dbo:award res:award-Q586356 .
res:person-Q25191 dbo:award res:award-Q787131 .
res:person-Q25191 dbo:award res:award-Q833154 .
res:person-Q25191 dbo:award res:award-Q833163 .
res:person-Q25191 ex:hasContribution res:contribution-Q13417189-Q25191-director .
res:person-Q25191 ex:hasContribution res:contribution-Q13417189-Q25191-producer .
res:person-Q25191 ex:hasContribution res:contribution-Q13417189-Q25191-writer .
res:person-Q25191 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:Person .
```

DBPEDIA VOCABULARY: dbo:Film, dbo:Person, dbo:Actor and the source-supported properties in Table 2.

ONTOLOGY RULE: (dbo:Person and ex:hasContribution min 3 ex:Contribution)

REASONING: The source relation supplies existential witnesses. Supporting credit classes follow role hasValue restrictions. For min 3, the three role individuals are distinct by their controlled meanings; functional hasRole forces the corresponding credits to be distinct. Hermit classified the complete graph.

INFERRED KNOWLEDGE: res:person-Q25191 rdf:type ex:ThreeCreditContributor. This type has zero direct assertions in the input graph.

SPARQL:

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a ex:ThreeCreditContributor .
  ↪ FILTER(STRSTARTS(STR(?s),
  ↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
  ↪ ?s rdfs:label ?label } } ORDER BY ?s
```

RESULT: 7 newly entailed local individuals; example Christopher Nolan (res:person-Q25191).

WHY THIS DEMONSTRATES SEMANTIC WEB: The ontology makes the classification and

shared vocabulary explicit and reusable across consumers. SQL joins, recursive queries and rule systems can reproduce many of these answers; the advantage is declared semantics, interoperable identifiers, open-world entailment and consistency checking, not an inability of databases to compute them.

Which action films received an award?

QUESTION: Which action films received an award?

INITIAL TRIPLES:

```

res:film-Q103569 dbo:award res:award-Q1131772 .
res:film-Q103569 dbo:award res:award-Q1257399 .
res:film-Q103569 dbo:award res:award-Q1265702 .
res:film-Q103569 dbo:award res:award-Q3414212 .
res:film-Q103569 dbo:award res:award-Q393686 .
res:film-Q103569 dbo:award res:award-Q739633 .
res:film-Q103569 dbo:genre res:genre-Q102260466 .
res:film-Q103569 dbo:genre res:genre-Q10663882 .
res:film-Q103569 dbo:genre res:genre-Q11304653 .
res:film-Q103569 dbo:genre res:genre-Q1200678 .
res:film-Q103569 dbo:genre res:genre-Q1342372 .
res:film-Q103569 dbo:genre res:genre-Q157394 .
res:film-Q103569 dbo:genre res:genre-Q1776156 .
res:film-Q103569 dbo:genre res:genre-Q188473 .
res:film-Q103569 dbo:genre res:genre-Q200092 .
res:film-Q103569 dbo:genre res:genre-Q20656232 .
res:film-Q103569 dbo:genre res:genre-Q2484376 .
res:film-Q103569 dbo:genre res:genre-Q319221 .
res:film-Q103569 dbo:genre res:genre-Q471839 .
res:film-Q103569 dbo:starring res:person-Q102124 .
res:film-Q103569 dbo:starring res:person-Q200405 .
res:film-Q103569 dbo:starring res:person-Q223091 .
res:film-Q103569 dbo:starring res:person-Q266270 .
res:film-Q103569 dbo:starring res:person-Q283872 .
res:film-Q103569 dbo:starring res:person-Q3047302 .
res:film-Q103569 dbo:starring res:person-Q314290 .
res:film-Q103569 dbo:starring res:person-Q320093 .
res:film-Q103569 dbo:starring res:person-Q323185 .
res:film-Q103569 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:Film .
res:genre-Q102260466 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q10663882 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q11304653 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q1200678 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q1342372 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q157394 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q1776156 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q188473 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q188473 http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:ActionGenre .
res:genre-Q200092 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q20656232 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q20656232 http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:ActionGenre
↪ .
res:genre-Q2484376 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q319221 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q471839 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .

```

DBPEDIA VOCABULARY: dbo:Film, dbo:Person, dbo:Actor and the source-supported proper-

ties in Table 2.

ONTOLOGY RULE: (ex:ActionFilm and ex:AwardWinningFilm)

REASONING: The source relation supplies existential witnesses. Supporting credit classes follow role hasValue restrictions. For min 3, the three role individuals are distinct by their controlled meanings; functional hasRole forces the corresponding credits to be distinct. Hermit classified the complete graph.

INFERRED KNOWLEDGE: res:film-Q103569 rdf:type ex:AwardWinningActionFilm. This type has zero direct assertions in the input graph.

SPARQL:

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a ex:AwardWinningActionFilm .
  ↪ FILTER(STRSTARTS(STR(?s),
  ↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
  ↪ ?s rdfs:label ?label } } ORDER BY ?s
```

RESULT: 10 newly entailed local individuals; example Alien (res:film-Q103569).

WHY THIS DEMONSTRATES SEMANTIC WEB: The ontology makes the classification and shared vocabulary explicit and reusable across consumers. SQL joins, recursive queries and rule systems can reproduce many of these answers; the advantage is declared semantics, interoperable identifiers, open-world entailment and consistency checking, not an inability of databases to compute them.

10. Full definitions, source witnesses and OWL expressions

ex:ActingContribution

REUSE/EXTEND rationale: Local credit record classified by a fixed role; DBpedia Actor describes a person, not this record.

CRAWLED SUPPORT: Credit claim corresponding to Actor

MANCHESTER: (ex:Contribution and ex:hasRole value ex:ActorRole)

OWL EXPRESSION: EquivalentClasses(ex:ActingContribution ObjectIntersectionOf(ex:Contribution ObjectHasValue(ex:hasRole ex:ActorRole)))

REAL INDIVIDUAL: Alien – actor – Sigourney Weaver (res:contribution-Q103569-Q102124-actor).

ASSERTED / INFERRED: No direct assertion of this type; Hermit entails membership. The role→credit→person classes need approximately 2–4 explanatory steps; OWL semantics is not procedural and does not prescribe evaluation order.

INITIAL FACTS:

```
res:contribution-Q103569-Q102124-actor ex:hasRole ex:ActorRole .
res:contribution-Q103569-Q102124-actor
  ↪ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
```

NEW KNOWLEDGE: res:contribution-Q103569-Q102124-actor rdf:type ex:ActingContribution.

ex:DirectingContribution

REUSE/EXTEND rationale: Local credit record classified by a fixed role; DBpedia Actor describes a person, not this record.

CRAWLED SUPPORT: Credit claim corresponding to Director

MANCHESTER: (ex:Contribution and ex:hasRole value ex:DirectorRole)

OWL EXPRESSION: EquivalentClasses(ex:DirectingContribution ObjectIntersectionOf(ex:Contribution ObjectHasValue(ex:hasRole ex:DirectorRole)))

REAL INDIVIDUAL: Alien – director – Ridley Scott (res:contribution-Q103569-Q56005-director).

ASSERTED / INFERRED: No direct assertion of this type; HermiT entails membership. The role→credit→person classes need approximately 2–4 explanatory steps; OWL semantics is not procedural and does not prescribe evaluation order.

INITIAL FACTS:

```
res:contribution-Q103569-Q56005-director ex:hasRole ex:DirectorRole .
res:contribution-Q103569-Q56005-director
  ↪ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
```

NEW KNOWLEDGE: res:contribution-Q103569-Q56005-director rdf:type ex:DirectingContribution.

ex:WritingContribution

REUSE/EXTEND rationale: Local credit record classified by a fixed role; DBpedia Actor describes a person, not this record.

CRAWLED SUPPORT: Credit claim corresponding to Writer

MANCHESTER: (ex:Contribution and ex:hasRole value ex:WriterRole)

OWL EXPRESSION: EquivalentClasses(ex:WritingContribution ObjectIntersectionOf(ex:Contribution ObjectHasValue(ex:hasRole ex:WriterRole)))

REAL INDIVIDUAL: Alien – writer – Ronald Shusett (res:contribution-Q103569-Q3056094-writer).

ASSERTED / INFERRED: No direct assertion of this type; HermiT entails membership. The role→credit→person classes need approximately 2–4 explanatory steps; OWL semantics is not procedural and does not prescribe evaluation order.

INITIAL FACTS:

```
res:contribution-Q103569-Q3056094-writer ex:hasRole ex:WriterRole .
res:contribution-Q103569-Q3056094-writer
  ↪ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
```

NEW KNOWLEDGE: res:contribution-Q103569-Q3056094-writer rdf:type ex:WritingContribution.

ex:ProducingContribution

REUSE/EXTEND rationale: Local credit record classified by a fixed role; DBpedia Actor describes a person, not this record.

CRAWLED SUPPORT: Credit claim corresponding to Producer

MANCHESTER: (ex:Contribution and ex:hasRole value ex:ProducerRole)

OWL EXPRESSION: EquivalentClasses(ex:ProducingContribution ObjectIntersectionOf(ex:Contribution ObjectHasValue(ex:hasRole ex:ProducerRole)))

REAL INDIVIDUAL: Alien – producer – Gordon Carroll (res:contribution-Q103569-Q1537983-producer).

ASSERTED / INFERRED: No direct assertion of this type; HermiT entails membership. The role→credit→person classes need approximately 2–4 explanatory steps; OWL semantics is not procedural and does not prescribe evaluation order.

INITIAL FACTS:

res:contribution-Q103569-Q1537983-producer ex:hasRole ex:ProducerRole .
 res:contribution-Q103569-Q1537983-producer
 ⇨ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .

NEW KNOWLEDGE: res:contribution-Q103569-Q1537983-producer rdf:type ex:ProducingContribution.

ex:Filmmaker

REUSE/EXTEND rationale: Local dataset-specific OWL-defined subset; not a replacement for a DBpedia base class.

CRAWLED SUPPORT: P57/P161/P58/P162; P166 when award-based

MANCHESTER: (dbo:Person and (ex:hasContribution some ex:DirectingContribution or ex:hasContribution some ex:WritingContribution or ex:hasContribution some ex:ProducingContribution))

OWL EXPRESSION: EquivalentClasses(ex:Filmmaker ObjectIntersectionOf(dbo:Person ObjectUnionOf(ObjectSomeValuesFrom(ex:hasContribution ex:DirectingContribution) ObjectSomeValuesFrom(ex:hasContribution ex:WritingContribution) ObjectSomeValuesFrom(ex:hasContribution ex:ProducingContribution))))

REAL INDIVIDUAL: Philippa Boyens (res:person-Q116854).

ASSERTED / INFERRED: No direct assertion of this type; HermiT entails membership. The role→credit→person classes need approximately 2–4 explanatory steps; OWL semantics is not procedural and does not prescribe evaluation order.

INITIAL FACTS:

res:contribution-Q127367-Q116854-writer ex:contributionTo res:film-Q127367 .
 res:contribution-Q127367-Q116854-writer ex:hasRole ex:WriterRole .
 res:contribution-Q127367-Q116854-writer
 ⇨ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
 res:person-Q116854 dbo:award res:award-Q1056240 .
 res:person-Q116854 dbo:award res:award-Q107258 .
 res:person-Q116854 dbo:award res:award-Q1075366 .
 res:person-Q116854 dbo:award res:award-Q23050104 .
 res:person-Q116854 dbo:award res:award-Q3414212 .
 res:person-Q116854 dbo:award res:award-Q430910 .
 res:person-Q116854 dbo:award res:award-Q644077 .
 res:person-Q116854 dbo:award res:award-Q739694 .
 res:person-Q116854 ex:hasContribution res:contribution-Q127367-Q116854-writer .
 res:person-Q116854 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:Person .

NEW KNOWLEDGE: res:person-Q116854 rdf:type ex:Filmmaker.

ex:ActionFilm

REUSE/EXTEND rationale: Local dataset-specific OWL-defined subset; not a replacement for a DBpedia base class.

CRAWLED SUPPORT: P136/P166

MANCHESTER: (dbo:Film and dbo:genre some ex:ActionGenre)

OWL EXPRESSION: EquivalentClasses(ex:ActionFilm ObjectIntersectionOf(dbo:Film ObjectSomeValuesFrom(dbo:genre ex:ActionGenre)))

REAL INDIVIDUAL: Alien (res:film-Q103569).

ASSERTED / INFERRED: No direct assertion of this type; HermiT entails membership. The role→credit→person classes need approximately 2–4 explanatory steps; OWL semantics is not procedural and does not prescribe evaluation order.

INITIAL FACTS:

```

res:film-Q103569 dbo:award res:award-Q1131772 .
res:film-Q103569 dbo:award res:award-Q1257399 .
res:film-Q103569 dbo:award res:award-Q1265702 .
res:film-Q103569 dbo:award res:award-Q3414212 .
res:film-Q103569 dbo:award res:award-Q393686 .
res:film-Q103569 dbo:award res:award-Q739633 .
res:film-Q103569 dbo:genre res:genre-Q102260466 .
res:film-Q103569 dbo:genre res:genre-Q10663882 .
res:film-Q103569 dbo:genre res:genre-Q11304653 .
res:film-Q103569 dbo:genre res:genre-Q1200678 .
res:film-Q103569 dbo:genre res:genre-Q1342372 .
res:film-Q103569 dbo:genre res:genre-Q157394 .
res:film-Q103569 dbo:genre res:genre-Q1776156 .
res:film-Q103569 dbo:genre res:genre-Q188473 .
res:film-Q103569 dbo:genre res:genre-Q200092 .
res:film-Q103569 dbo:genre res:genre-Q20656232 .
res:film-Q103569 dbo:genre res:genre-Q2484376 .
res:film-Q103569 dbo:genre res:genre-Q319221 .
res:film-Q103569 dbo:genre res:genre-Q471839 .
res:film-Q103569 dbo:starring res:person-Q102124 .
res:film-Q103569 dbo:starring res:person-Q200405 .
res:film-Q103569 dbo:starring res:person-Q223091 .
res:film-Q103569 dbo:starring res:person-Q266270 .
res:film-Q103569 dbo:starring res:person-Q283872 .
res:film-Q103569 dbo:starring res:person-Q3047302 .
res:film-Q103569 dbo:starring res:person-Q314290 .
res:film-Q103569 dbo:starring res:person-Q320093 .
res:film-Q103569 dbo:starring res:person-Q323185 .
res:film-Q103569 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:Film .
res:genre-Q102260466 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q10663882 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q11304653 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q1200678 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q1342372 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q157394 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q1776156 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q188473 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q188473 http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:ActionGenre .
res:genre-Q200092 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q20656232 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q20656232 http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:ActionGenre
↪ .
res:genre-Q2484376 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q319221 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q471839 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .

```

NEW KNOWLEDGE: res:film-Q103569 rdf:type ex:ActionFilm.

ex:AwardWinningFilm

REUSE/EXTEND rationale: Local dataset-specific OWL-defined subset; not a replacement for a DBpedia base class.

CRAWLED SUPPORT: P136/P166

MANCHESTER: (dbo:Film and dbo:award some dbo:Award)

OWL EXPRESSION: `EquivalentClasses(ex:AwardWinningFilm ObjectIntersectionOf(dbo:Film ObjectSomeValuesFrom(dbo:award dbo:Award)))`

REAL INDIVIDUAL: `Alien (res:film-Q103569)`.

ASSERTED / INFERRED: No direct assertion of this type; Hermit entails membership. The role `→credit→person` classes need approximately 2–4 explanatory steps; OWL semantics is not procedural and does not prescribe evaluation order.

INITIAL FACTS:

```
res:film-Q103569 dbo:award res:award-Q1131772 .
res:film-Q103569 dbo:award res:award-Q1257399 .
res:film-Q103569 dbo:award res:award-Q1265702 .
res:film-Q103569 dbo:award res:award-Q3414212 .
res:film-Q103569 dbo:award res:award-Q393686 .
res:film-Q103569 dbo:award res:award-Q739633 .
res:film-Q103569 dbo:genre res:genre-Q102260466 .
res:film-Q103569 dbo:genre res:genre-Q10663882 .
res:film-Q103569 dbo:genre res:genre-Q11304653 .
res:film-Q103569 dbo:genre res:genre-Q1200678 .
res:film-Q103569 dbo:genre res:genre-Q1342372 .
res:film-Q103569 dbo:genre res:genre-Q157394 .
res:film-Q103569 dbo:genre res:genre-Q1776156 .
res:film-Q103569 dbo:genre res:genre-Q188473 .
res:film-Q103569 dbo:genre res:genre-Q200092 .
res:film-Q103569 dbo:genre res:genre-Q20656232 .
res:film-Q103569 dbo:genre res:genre-Q2484376 .
res:film-Q103569 dbo:genre res:genre-Q319221 .
res:film-Q103569 dbo:genre res:genre-Q471839 .
res:film-Q103569 dbo:starring res:person-Q102124 .
res:film-Q103569 dbo:starring res:person-Q200405 .
res:film-Q103569 dbo:starring res:person-Q223091 .
res:film-Q103569 dbo:starring res:person-Q266270 .
res:film-Q103569 dbo:starring res:person-Q283872 .
res:film-Q103569 dbo:starring res:person-Q3047302 .
res:film-Q103569 dbo:starring res:person-Q314290 .
res:film-Q103569 dbo:starring res:person-Q320093 .
res:film-Q103569 dbo:starring res:person-Q323185 .
res:film-Q103569 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:Film .
res:genre-Q102260466 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q10663882 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q11304653 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q1200678 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q1342372 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q157394 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q1776156 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q188473 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q188473 http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:ActionGenre .
res:genre-Q200092 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q20656232 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q20656232 http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:ActionGenre
↪ .
res:genre-Q2484376 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q319221 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q471839 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
```

NEW KNOWLEDGE: `res:film-Q103569 rdf:type ex:AwardWinningFilm`.

ex:MultiCreditContributor

REUSE/EXTEND rationale: At least two semantically distinct credit records. Different roles prove their distinctness; the name is shorthand and does not exclude repeated same-role credits.

CRAWLED SUPPORT: P57/P161/P58/P162; P166 when award-based

MANCHESTER: (dbo:Person and ex:hasContribution min 2 ex:Contribution)

OWL EXPRESSION: `EquivalentClasses(ex:MultiCreditContributor ObjectIntersectionOf(dbo:Person ObjectMinCardinality(2 ex:hasContribution ex:Contribution)))`

REAL INDIVIDUAL: Fran Walsh (res:person-Q116861).

ASSERTED / INFERRED: No direct assertion of this type; HermiT entails membership. The role→credit→person classes need approximately 2–4 explanatory steps; OWL semantics is not procedural and does not prescribe evaluation order.

INITIAL FACTS:

```
res:contribution-Q127367-Q116861-producer ex:contributionTo res:film-Q127367 .
res:contribution-Q127367-Q116861-producer ex:hasRole ex:ProducerRole .
res:contribution-Q127367-Q116861-producer
  ↪ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:contribution-Q127367-Q116861-writer ex:contributionTo res:film-Q127367 .
res:contribution-Q127367-Q116861-writer ex:hasRole ex:WriterRole .
res:contribution-Q127367-Q116861-writer
  ↪ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:person-Q116861 dbo:award res:award-Q102427 .
res:person-Q116861 dbo:award res:award-Q1056240 .
res:person-Q116861 dbo:award res:award-Q107258 .
res:person-Q116861 dbo:award res:award-Q1075366 .
res:person-Q116861 dbo:award res:award-Q112243 .
res:person-Q116861 dbo:award res:award-Q131837474 .
res:person-Q116861 dbo:award res:award-Q139184 .
res:person-Q116861 dbo:award res:award-Q1472235 .
res:person-Q116861 dbo:award res:award-Q23049782 .
res:person-Q116861 dbo:award res:award-Q23050104 .
res:person-Q116861 dbo:award res:award-Q3414212 .
res:person-Q116861 dbo:award res:award-Q428808 .
res:person-Q116861 dbo:award res:award-Q430910 .
res:person-Q116861 dbo:award res:award-Q4824149 .
res:person-Q116861 dbo:award res:award-Q5569374 .
res:person-Q116861 dbo:award res:award-Q644077 .
res:person-Q116861 dbo:award res:award-Q739694 .
res:person-Q116861 ex:hasContribution res:contribution-Q127367-Q116861-producer .
res:person-Q116861 ex:hasContribution res:contribution-Q127367-Q116861-writer .
res:person-Q116861 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:Person .
```

NEW KNOWLEDGE: res:person-Q116861 rdf:type ex:MultiCreditContributor.

ex:ThreeCreditContributor

REUSE/EXTEND rationale: At least three distinct credit records; three different role fillers prove distinctness. Not exactly three roles or an exhaustive career description.

CRAWLED SUPPORT: P57/P161/P58/P162; P166 when award-based

MANCHESTER: (dbo:Person and ex:hasContribution min 3 ex:Contribution)

OWL EXPRESSION: `EquivalentClasses(ex:ThreeCreditContributor ObjectIntersectionOf(dbo:Person ObjectMinCardinality(3 ex:hasContribution ex:Contribution)))`

REAL INDIVIDUAL: Christopher Nolan (res:person-Q25191).

ASSERTED / INFERRED: No direct assertion of this type; HermiT entails membership. The

role→credit→person classes need approximately 2–4 explanatory steps; OWL semantics is not procedural and does not prescribe evaluation order.

INITIAL FACTS:

```
res:contribution-Q13417189-Q25191-director ex:contributionTo res:film-Q13417189 .
res:contribution-Q13417189-Q25191-director ex:hasRole ex:DirectorRole .
res:contribution-Q13417189-Q25191-director
  ⇨ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:contribution-Q13417189-Q25191-producer ex:contributionTo res:film-Q13417189 .
res:contribution-Q13417189-Q25191-producer ex:hasRole ex:ProducerRole .
res:contribution-Q13417189-Q25191-producer
  ⇨ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:contribution-Q13417189-Q25191-writer ex:contributionTo res:film-Q13417189 .
res:contribution-Q13417189-Q25191-writer ex:hasRole ex:WriterRole .
res:contribution-Q13417189-Q25191-writer
  ⇨ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:person-Q25191 dbo:award res:award-Q102427 .
res:person-Q25191 dbo:award res:award-Q103360 .
res:person-Q25191 dbo:award res:award-Q105447 .
res:person-Q25191 dbo:award res:award-Q1056240 .
res:person-Q25191 dbo:award res:award-Q12143912 .
res:person-Q25191 dbo:award res:award-Q12201477 .
res:person-Q25191 dbo:award res:award-Q16058279 .
res:person-Q25191 dbo:award res:award-Q3405197 .
res:person-Q25191 dbo:award res:award-Q586356 .
res:person-Q25191 dbo:award res:award-Q787131 .
res:person-Q25191 dbo:award res:award-Q833154 .
res:person-Q25191 dbo:award res:award-Q833163 .
res:person-Q25191 ex:hasContribution res:contribution-Q13417189-Q25191-director .
res:person-Q25191 ex:hasContribution res:contribution-Q13417189-Q25191-producer .
res:person-Q25191 ex:hasContribution res:contribution-Q13417189-Q25191-writer .
res:person-Q25191 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:Person .
```

NEW KNOWLEDGE: res:person-Q25191 rdf:type ex:ThreeCreditContributor.

ex:WriterDirector

REUSE/EXTEND rationale: A person with both directing and writing credits in this dataset; classification is never manually asserted.

CRAWLED SUPPORT: P57/P161/P58/P162; P166 when award-based

MANCHESTER: (dbo:MovieDirector and dbo:ScreenWriter)

OWL EXPRESSION: EquivalentClasses(ex:WriterDirector ObjectIntersectionOf(dbo:MovieDirector dbo:ScreenWriter))

REAL INDIVIDUAL: Damien Chazelle (res:person-Q18350026).

ASSERTED / INFERRED: No direct assertion of this type; Hermit entails membership. The role→credit→person classes need approximately 2–4 explanatory steps; OWL semantics is not procedural and does not prescribe evaluation order.

INITIAL FACTS:

```
res:contribution-Q15648198-Q18350026-director ex:contributionTo res:film-Q15648198
  ⇨ .
res:contribution-Q15648198-Q18350026-director ex:hasRole ex:DirectorRole .
res:contribution-Q15648198-Q18350026-director
  ⇨ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:contribution-Q15648198-Q18350026-writer ex:contributionTo res:film-Q15648198 .
res:contribution-Q15648198-Q18350026-writer ex:hasRole ex:WriterRole .
res:contribution-Q15648198-Q18350026-writer
  ⇨ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
```

```

res:person-Q18350026 dbo:award res:award-Q103360 .
res:person-Q18350026 dbo:award res:award-Q3774974 .
res:person-Q18350026 dbo:award res:award-Q787131 .
res:person-Q18350026 ex:hasContribution
  ↪ res:contribution-Q15648198-Q18350026-director .
res:person-Q18350026 ex:hasContribution res:contribution-Q15648198-Q18350026-writer
  ↪ .
res:person-Q18350026 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:Person .
NEW KNOWLEDGE: res:person-Q18350026 rdf:type ex:WriterDirector.

```

ex:ActorFilmmaker

REUSE/EXTEND rationale: Local dataset-specific OWL-defined subset; not a replacement for a DBpedia base class.

CRAWLED SUPPORT: P57/P161/P58/P162; P166 when award-based

MANCHESTER: (dbo:Actor and ex:Filmmaker)

OWL EXPRESSION: EquivalentClasses(ex:ActorFilmmaker ObjectIntersectionOf(dbo:Actor ex:Filmmaker))

REAL INDIVIDUAL: Quentin Tarantino (res:person-Q3772).

ASSERTED / INFERRED: No direct assertion of this type; Hermit entails membership. The role→credit→person classes need approximately 2–4 explanatory steps; OWL semantics is not procedural and does not prescribe evaluation order.

INITIAL FACTS:

```

res:contribution-Q104123-Q3772-actor ex:contributionTo res:film-Q104123 .
res:contribution-Q104123-Q3772-actor ex:hasRole ex:ActorRole .
res:contribution-Q104123-Q3772-actor
  ↪ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:contribution-Q104123-Q3772-director ex:contributionTo res:film-Q104123 .
res:contribution-Q104123-Q3772-director ex:hasRole ex:DirectorRole .
res:contribution-Q104123-Q3772-director
  ↪ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:contribution-Q104123-Q3772-writer ex:contributionTo res:film-Q104123 .
res:contribution-Q104123-Q3772-writer ex:hasRole ex:WriterRole .
res:contribution-Q104123-Q3772-writer
  ↪ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:film-Q104123 dbo:starring res:person-Q3772 .
res:person-Q3772 dbo:award res:award-Q1011547 .
res:person-Q3772 dbo:award res:award-Q105447 .
res:person-Q3772 dbo:award res:award-Q113068 .
res:person-Q3772 dbo:award res:award-Q13452531 .
res:person-Q3772 dbo:award res:award-Q1789102 .
res:person-Q3772 dbo:award res:award-Q179808 .
res:person-Q3772 dbo:award res:award-Q17985761 .
res:person-Q3772 dbo:award res:award-Q24051550 .
res:person-Q3772 dbo:award res:award-Q25405526 .
res:person-Q3772 dbo:award res:award-Q311836 .
res:person-Q3772 dbo:award res:award-Q3404975 .
res:person-Q3772 dbo:award res:award-Q3703457 .
res:person-Q3772 dbo:award res:award-Q41375 .
res:person-Q3772 dbo:award res:award-Q41417 .
res:person-Q3772 dbo:award res:award-Q4743376 .
res:person-Q3772 dbo:award res:award-Q5211225 .
res:person-Q3772 dbo:award res:award-Q52382875 .
res:person-Q3772 dbo:award res:award-Q670282 .
res:person-Q3772 dbo:award res:award-Q716909 .
res:person-Q3772 dbo:award res:award-Q727282 .

```

```

res:person-Q3772 dbo:award res:award-Q833154 .
res:person-Q3772 dbo:award res:award-Q849124 .
res:person-Q3772 dbo:award res:award-Q849771 .
res:person-Q3772 ex:hasContribution res:contribution-Q104123-Q3772-actor .
res:person-Q3772 ex:hasContribution res:contribution-Q104123-Q3772-director .
res:person-Q3772 ex:hasContribution res:contribution-Q104123-Q3772-writer .
res:person-Q3772 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:Person .
NEW KNOWLEDGE: res:person-Q3772 rdf:type ex:ActorFilmmaker.

```

ex:AwardWinningFilmmaker

REUSE/EXTEND rationale: Local dataset-specific OWL-defined subset; not a replacement for a DBpedia base class.

CRAWLED SUPPORT: P57/P161/P58/P162; P166 when award-based

MANCHESTER: (ex:Filmmaker and dbo:award some dbo:Award)

OWL EXPRESSION: `EquivalentClasses(ex:AwardWinningFilmmaker ObjectIntersectionOf(ex:Filmmaker ObjectSomeValuesFrom(dbo:award dbo:Award)))`

REAL INDIVIDUAL: Philippa Boyens (res:person-Q116854).

ASSERTED / INFERRED: No direct assertion of this type; Hermit entails membership. The role→credit→person classes need approximately 2–4 explanatory steps; OWL semantics is not procedural and does not prescribe evaluation order.

INITIAL FACTS:

```

res:contribution-Q127367-Q116854-writer ex:contributionTo res:film-Q127367 .
res:contribution-Q127367-Q116854-writer ex:hasRole ex:WriterRole .
res:contribution-Q127367-Q116854-writer
  ↪ http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:Contribution .
res:person-Q116854 dbo:award res:award-Q1056240 .
res:person-Q116854 dbo:award res:award-Q107258 .
res:person-Q116854 dbo:award res:award-Q1075366 .
res:person-Q116854 dbo:award res:award-Q23050104 .
res:person-Q116854 dbo:award res:award-Q3414212 .
res:person-Q116854 dbo:award res:award-Q430910 .
res:person-Q116854 dbo:award res:award-Q644077 .
res:person-Q116854 dbo:award res:award-Q739694 .
res:person-Q116854 ex:hasContribution res:contribution-Q127367-Q116854-writer .
res:person-Q116854 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:Person .
NEW KNOWLEDGE: res:person-Q116854 rdf:type ex:AwardWinningFilmmaker.

```

ex:AwardWinningActionFilm

REUSE/EXTEND rationale: Local dataset-specific OWL-defined subset; not a replacement for a DBpedia base class.

CRAWLED SUPPORT: P136/P166

MANCHESTER: (ex:ActionFilm and ex:AwardWinningFilm)

OWL EXPRESSION: `EquivalentClasses(ex:AwardWinningActionFilm ObjectIntersectionOf(ex:ActionFilm ex:AwardWinningFilm))`

REAL INDIVIDUAL: Alien (res:film-Q103569).

ASSERTED / INFERRED: No direct assertion of this type; Hermit entails membership. The role→credit→person classes need approximately 2–4 explanatory steps; OWL semantics is not procedural and does not prescribe evaluation order.

INITIAL FACTS:


```

res:film-Q103569 dbo:award res:award-Q1131772 .
res:film-Q103569 dbo:award res:award-Q1257399 .
res:film-Q103569 dbo:award res:award-Q1265702 .
res:film-Q103569 dbo:award res:award-Q3414212 .
res:film-Q103569 dbo:award res:award-Q393686 .
res:film-Q103569 dbo:award res:award-Q739633 .
res:film-Q103569 dbo:genre res:genre-Q102260466 .
res:film-Q103569 dbo:genre res:genre-Q10663882 .
res:film-Q103569 dbo:genre res:genre-Q11304653 .
res:film-Q103569 dbo:genre res:genre-Q1200678 .
res:film-Q103569 dbo:genre res:genre-Q1342372 .
res:film-Q103569 dbo:genre res:genre-Q157394 .
res:film-Q103569 dbo:genre res:genre-Q1776156 .
res:film-Q103569 dbo:genre res:genre-Q188473 .
res:film-Q103569 dbo:genre res:genre-Q200092 .
res:film-Q103569 dbo:genre res:genre-Q20656232 .
res:film-Q103569 dbo:genre res:genre-Q2484376 .
res:film-Q103569 dbo:genre res:genre-Q319221 .
res:film-Q103569 dbo:genre res:genre-Q471839 .
res:film-Q103569 dbo:starring res:person-Q102124 .
res:film-Q103569 dbo:starring res:person-Q200405 .
res:film-Q103569 dbo:starring res:person-Q223091 .
res:film-Q103569 dbo:starring res:person-Q266270 .
res:film-Q103569 dbo:starring res:person-Q283872 .
res:film-Q103569 dbo:starring res:person-Q3047302 .
res:film-Q103569 dbo:starring res:person-Q314290 .
res:film-Q103569 dbo:starring res:person-Q320093 .
res:film-Q103569 dbo:starring res:person-Q323185 .
res:film-Q103569 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:Film .
res:genre-Q102260466 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q10663882 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q11304653 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q1200678 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q1342372 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q157394 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q1776156 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q188473 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q188473 http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:ActionGenre .
res:genre-Q200092 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q20656232 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q20656232 http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:ActionGenre
↪ .
res:genre-Q2484376 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q319221 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q471839 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
NEW KNOWLEDGE: res:film-Q103569 rdf:type ex:AwardWinningActionFilm.

```

ex:GenreCrossingFilm

REUSE/EXTEND rationale: Film carrying both action and drama genre memberships; this is not an unproved two-distinct-genre cardinality claim.

CRAWLED SUPPORT: P136/P166

MANCHESTER: (dbo:Film and dbo:genre some ex:ActionGenre and dbo:genre some ex:DramaGenre)

OWL EXPRESSION: `EquivalentClasses(ex:GenreCrossingFilm ObjectIntersectionOf(dbo:Film ObjectSomeValuesFrom(dbo:genre ex:ActionGenre) ObjectSomeValuesFrom(dbo:genre ex:DramaGenre)))`

REAL INDIVIDUAL: `Pulp Fiction (res:film-Q104123)`.

ASSERTED / INFERRED: No direct assertion of this type; Hermit entails membership. The role→credit→person classes need approximately 2–4 explanatory steps; OWL semantics is not procedural and does not prescribe evaluation order.

INITIAL FACTS:

```
res:film-Q104123 dbo:award res:award-Q1170507 .
res:film-Q104123 dbo:award res:award-Q1315008 .
res:film-Q104123 dbo:award res:award-Q1720784 .
res:film-Q104123 dbo:award res:award-Q1789102 .
res:film-Q104123 dbo:award res:award-Q179808 .
res:film-Q104123 dbo:award res:award-Q1868950 .
res:film-Q104123 dbo:award res:award-Q1966965 .
res:film-Q104123 dbo:award res:award-Q2295041 .
res:film-Q104123 dbo:award res:award-Q2544859 .
res:film-Q104123 dbo:award res:award-Q3404523 .
res:film-Q104123 dbo:award res:award-Q3841643 .
res:film-Q104123 dbo:award res:award-Q41375 .
res:film-Q104123 dbo:award res:award-Q41417 .
res:film-Q104123 dbo:award res:award-Q503034 .
res:film-Q104123 dbo:award res:award-Q505449 .
res:film-Q104123 dbo:award res:award-Q5211226 .
res:film-Q104123 dbo:award res:award-Q540977 .
res:film-Q104123 dbo:award res:award-Q548389 .
res:film-Q104123 dbo:award res:award-Q696972 .
res:film-Q104123 dbo:award res:award-Q696998 .
res:film-Q104123 dbo:award res:award-Q6978541 .
res:film-Q104123 dbo:award res:award-Q849124 .
res:film-Q104123 dbo:award res:award-Q952914 .
res:film-Q104123 dbo:genre res:genre-Q11304653 .
res:film-Q104123 dbo:genre res:genre-Q113485322 .
res:film-Q104123 dbo:genre res:genre-Q130232 .
res:film-Q104123 dbo:genre res:genre-Q157443 .
res:film-Q104123 dbo:genre res:genre-Q1788980 .
res:film-Q104123 dbo:genre res:genre-Q188473 .
res:film-Q104123 dbo:genre res:genre-Q19367312 .
res:film-Q104123 dbo:genre res:genre-Q2421031 .
res:film-Q104123 dbo:genre res:genre-Q2484376 .
res:film-Q104123 dbo:genre res:genre-Q3990883 .
res:film-Q104123 dbo:genre res:genre-Q459290 .
res:film-Q104123 dbo:genre res:genre-Q5778924 .
res:film-Q104123 dbo:genre res:genre-Q7444356 .
res:film-Q104123 dbo:genre res:genre-Q859369 .
res:film-Q104123 dbo:genre res:genre-Q959790 .
res:film-Q104123 dbo:starring res:person-Q1016897 .
res:film-Q104123 dbo:starring res:person-Q104061 .
res:film-Q104123 dbo:starring res:person-Q109232 .
res:film-Q104123 dbo:starring res:person-Q1209753 .
res:film-Q104123 dbo:starring res:person-Q125017 .
res:film-Q104123 dbo:starring res:person-Q1368224 .
res:film-Q104123 dbo:starring res:person-Q1563838 .
res:film-Q104123 dbo:starring res:person-Q164534 .
res:film-Q104123 dbo:starring res:person-Q172678 .
res:film-Q104123 dbo:starring res:person-Q185051 .
res:film-Q104123 dbo:starring res:person-Q191132 .
res:film-Q104123 dbo:starring res:person-Q203804 .
res:film-Q104123 dbo:starring res:person-Q2263306 .
```

```

res:film-Q104123 dbo:starring res:person-Q235115 .
res:film-Q104123 dbo:starring res:person-Q238483 .
res:film-Q104123 dbo:starring res:person-Q240774 .
res:film-Q104123 dbo:starring res:person-Q2434817 .
res:film-Q104123 dbo:starring res:person-Q254431 .
res:film-Q104123 dbo:starring res:person-Q270774 .
res:film-Q104123 dbo:starring res:person-Q2924399 .
res:film-Q104123 dbo:starring res:person-Q2926038 .
res:film-Q104123 dbo:starring res:person-Q310315 .
res:film-Q104123 dbo:starring res:person-Q318267 .
res:film-Q104123 dbo:starring res:person-Q3193117 .
res:film-Q104123 dbo:starring res:person-Q3241353 .
res:film-Q104123 dbo:starring res:person-Q356541 .
res:film-Q104123 dbo:starring res:person-Q3772 .
res:film-Q104123 dbo:starring res:person-Q3856171 .
res:film-Q104123 dbo:starring res:person-Q3973186 .
res:film-Q104123 dbo:starring res:person-Q432437 .
res:film-Q104123 dbo:starring res:person-Q453973 .
res:film-Q104123 dbo:starring res:person-Q514913 .
res:film-Q104123 dbo:starring res:person-Q7647898 .
res:film-Q104123 dbo:starring res:person-Q80938 .
res:film-Q104123 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:Film .
res:genre-Q11304653 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q113485322 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q113485322 http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:DramaGenre
↪ .
res:genre-Q130232 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q130232 http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:DramaGenre .
res:genre-Q157443 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q1788980 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q188473 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q188473 http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:ActionGenre .
res:genre-Q19367312 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre
↪ .
res:genre-Q2421031 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q2484376 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q3990883 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q3990883 http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:ActionGenre .
res:genre-Q459290 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q5778924 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q7444356 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q859369 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
res:genre-Q859369 http://www.w3.org/1999/02/22-rdf-syntax-ns#type ex:DramaGenre .
res:genre-Q959790 http://www.w3.org/1999/02/22-rdf-syntax-ns#type dbo:MovieGenre .
NEW KNOWLEDGE: res:film-Q104123 rdf:type ex:GenreCrossingFilm.

```

11. Cardinality and validation limitations

Exactly 1 constrains each Contribution endpoint and role. It does not validate completeness under the open-world assumption: an absent filler may exist without an explicit assertion. Structural checks must separately enforce that the exported record contains its three fields.

At least 2 / at least 3 are genuine HermiT results on hasContribution. Example: Christopher Nolan has distinct directing, writing and producing credit records; those records cannot collapse into one because hasRole is functional and DirectorRole, WriterRole and ProducerRole are distinct. No inequality is fabricated for people, genres, awards, companies or films.

The negative MultiGenreFilm test separately adds the requested min 2 genre definition and

obtains zero DL members. A production class is excluded because the dataset cannot prove two genre fillers distinct. Future additions require source-supported inequality or a justified identity model; a blanket AllDifferent on Wikidata IDs is not justified by syntax.

12. Checklist and reproduction

- ☒ Official DBpedia reference checked and hashed; reused term meanings, domain/range preserved.
- ☒ No ex:Film, Person, Actor, Award, Genre, Language, Country, Company or Organisation clones.
- ☒ Every extension maps to crawled claims or documented source-record metadata.
- ☒ All demo individuals exist in the crawled dataset.
- ☒ All retained defined classes have genuinely inferred members.
- ☒ Equivalent classes, existential/value/cardinality restrictions and hierarchy present.
- ☒ Inverse, subproperty and chain reasoning present; no gratuitous property characteristics.
- ☒ Seven explanatory inference chains and 27 executed competency questions.
- ☒ Five verified demo cases with no false claim that SQL cannot reproduce them.
- ☒ Genuine min 2/min 3 DL results and explicit negative genre-cardinality test.
- ☒ No MP4 videos included; removed at user request.
- ☐ Zero-result MultiGenreFilm / MultiAwardFilm / FrequentProductionCompany: require additional justified evidence before inclusion.
- ☒ Archived official ontology, official development vocabulary and relevant live term meanings checked; equivalence decisions remain scoped to these references.

Run:

```
make build
make reason JAVA=/path/to/java
make validate
make test
```

Open ontology/Movie_Knowledge_Graph.owl directly in Protégé; it combines the schema and asserted facts without pre-asserting inferred classes. Start HermiT and inspect the example types; inferred.ttl is exported entailment evidence, not source assertions. before_after.trig supports query 27. Temporary RDF/XML reasoner inputs are removed after each run, preserving one authoritative full OWL export.

Property-by-property semantic reuse audit

The comparison checks the relation meaning, argument direction, entity level, range and units; lexical resemblance alone is insufficient.

Property	Decision	Semantic comparison
dbo:director	REUSE	Same relation/value semantics as the checked DBpedia property, with its original domain and range. Runtime is converted to the reference unit of seconds.
dbo:starring	REUSE	Same relation/value semantics as the checked DBpedia property, with its original domain and range. Runtime is converted to the reference unit of seconds.

Property	Decision	Semantic comparison
dbo:writer	REUSE	Same relation/value semantics as the checked DBpedia property, with its original domain and range. Runtime is converted to the reference unit of seconds.
dbo:producer	REUSE	Same relation/value semantics as the checked DBpedia property, with its original domain and range. Runtime is converted to the reference unit of seconds.
dbo:productionCompany	REUSE	Same relation/value semantics as the checked DBpedia property, with its original domain and range. Runtime is converted to the reference unit of seconds.
dbo:award	REUSE	Same relation/value semantics as the checked DBpedia property, with its original domain and range. Runtime is converted to the reference unit of seconds.
dbo:genre	REUSE	Same relation/value semantics as the checked DBpedia property, with its original domain and range. Runtime is converted to the reference unit of seconds.
dbo:country	REUSE	Same relation/value semantics as the checked DBpedia property, with its original domain and range. Runtime is converted to the reference unit of seconds.
dbo:language	REUSE	Same relation/value semantics as the checked DBpedia property, with its original domain and range. Runtime is converted to the reference unit of seconds.
ex:hasContribution	CREATE	dbo:occupation links a Person Function, not a film-credit record. The local property refers to n-ary source credits.
ex:contributionBy	CREATE	dbo:director/writer/starring refer from Work/Film to people; this relation originates from a credit record, so they are not semantically interchangeable.
ex:contributionTo	CREATE	The domain is a credit association, not an Agent or Work. This is the film endpoint of the n-ary model.

Property	Decision	Semantic comparison
ex:contributionOf	CREATE	Inverse of the film endpoint of the local n-ary association; not a Film-to-Person relation.
ex:hasRole	CREATE	dbo:role and dbo:artistFunction have string ranges; the controlled-role entity of a functional credit record has distinct object semantics.
ex:contributedTo	CREATE	Covers every collected credit role through a shared chain. director/starring/writer/producer are narrower and opposite-direction properties.
ex:directed	CREATE	Inverse of dbo:director; cannot reuse that forward predicate with its arguments reversed. It is also a subproperty of the broader local contributedTo.
ex:actedIn	CREATE	Inverse of dbo:starring; forward dbo:starring cannot denote the reversed relation.
ex:productionOf	CREATE	Inverse of dbo:productionCompany; preserving direction is necessary for semantic equivalence.
ex:sourceSnapshot	CREATE	More specific than prov:wasDerivedFrom: points to a locally saved response record with retrieval timestamp and content hash, not merely any source entity.
dbo:runtime	REUSE	Same relation/value semantics as the checked DBpedia property, with its original domain and range. Runtime is converted to the reference unit of seconds.
ex:releaseYear	CREATE	Derived earliest-release-year integer summary from P577. dbo:releaseDate requires a date; dbo:year is a generic gYear and does not express the earliest-release summary. No equivalence or invented full date is asserted.
ex:sourceUrl	CREATE	Literal lexical request endpoint stored in the snapshot manifest. DBpedia has no matching request-metadata datatype property in the checked vocabulary; prov:wasDerivedFrom is separately retained as an IRI relation.

Property	Decision	Semantic comparison
ex:retrievedAt	CREATE	Retrieval instant of a local response snapshot, not a film releaseDate, generic date, or creation time of the original remote entity.
ex:sha256	CREATE	SHA-256 checksum of the exact crawled response bytes, not a generic title, name, code or identifier of the represented film/person.

References

- [Official DBpedia ontology download](#)
- [dbo:starring](#)
- [dbo:runtime](#)
- [dbo:MovieGenre](#)
- [DBpedia Actor meaning](#)
- [W3C OWL 2 Structural Specification](#)
- [W3C OWL 2 Primer](#)

Appendix – All executable competency queries

Namespace identifiers in the tables expand using these prefixes:

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
```

Query 1 – List films

Before / after rows: 30 / 30

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a dbo:Film . FILTER(STRSTARTS(STR(?s),
↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
↪ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 2 – Inception: year, duration and director

Before / after rows: 1 / 1

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT ?title ?year ?runtimeSeconds ?director WHERE { res:film-Q25188 rdfs:label
↪ ?title ; ex:releaseYear ?year ; dbo:runtime ?runtimeSeconds ; dbo:director
↪ ?person . ?person rdfs:label ?director }
```

Query 3 – Directors of Inception

Before / after rows: 1 / 1

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { res:film-Q25188 dbo:director ?s .
  ↪ FILTER(STRSTARTS(STR(?s),
  ↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
  ↪ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 4 – Inception credits and roles

Before / after rows: 25 / 25

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT ?personName ?roleName WHERE { res:film-Q25188 ex:contributionOf ?credit .
  ↪ ?credit ex:contributionBy ?person ; ex:hasRole ?role . ?person rdfs:label
  ↪ ?personName . ?role rdfs:label ?roleName } ORDER BY ?roleName ?personName
```

Query 5 – Films directed by Nolan

Before / after rows: 8 / 8

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s dbo:director res:person-Q25191 .
  ↪ FILTER(STRSTARTS(STR(?s),
  ↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
  ↪ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 6 – Companies credited on Inception

Before / after rows: 4 / 4

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { res:film-Q25188 dbo:productionCompany ?s .
  ↪ FILTER(STRSTARTS(STR(?s),
  ↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
  ↪ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 7 – Awards of Inception

Before / after rows: 7 / 7

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
```



```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { res:film-Q25188 dbo:award ?s .
  ↳ FILTER(STRSTARTS(STR(?s),
  ↳ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
  ↳ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 8 – Runtime in seconds

Before / after rows: 1 / 1

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { res:film-Q25188 dbo:runtime ?s . OPTIONAL { ?s
  ↳ rdfs:label ?label } } ORDER BY ?s
```

Query 9 – Film genres through the parent class

Before / after rows: 0 / 75

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?f a dbo:Film ; dbo:genre ?s . ?s a dbo:Genre .
  ↳ FILTER(STRSTARTS(STR(?s),
  ↳ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
  ↳ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 10 – Persons through Agent hierarchy

Before / after rows: 0 / 851

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a dbo:Agent . FILTER EXISTS { ?s a dbo:Person }
  ↳ FILTER(STRSTARTS(STR(?s),
  ↳ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
  ↳ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 11 – Production companies

Before / after rows: 45 / 45

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a dbo:Company . FILTER(STRSTARTS(STR(?s),
  ↳ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
  ↳ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 12 – Film subclasses

Before / after rows: 0 / 4

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s rdfs:subClassOf+ dbo:Film . OPTIONAL { ?s
↪ rdfs:label ?label } } ORDER BY ?s
```

Query 13 – Actors through Artist hierarchy

Before / after rows: 0 / 769

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a dbo:Artist . FILTER EXISTS { ?s a dbo:Actor }
↪ FILTER(STRSTARTS(STR(?s),
↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
↪ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 14 – Inferred Actors

Before / after rows: 0 / 769

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a dbo:Actor . FILTER(STRSTARTS(STR(?s),
↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
↪ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 15 – Inferred Filmmakers

Before / after rows: 0 / 89

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a ex:Filmmaker . FILTER(STRSTARTS(STR(?s),
↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
↪ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 16 – Inferred Action films

Before / after rows: 0 / 12

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
```

```
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a ex:ActionFilm . FILTER(STRSTARTS(STR(?s),
↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
↪ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 17 – Award-winning films

Before / after rows: 0 / 26

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a ex:AwardWinningFilm .
↪ FILTER(STRSTARTS(STR(?s),
↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
↪ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 18 – People with at least two provably distinct credits

Before / after rows: 0 / 17

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a ex:MultiCreditContributor .
↪ FILTER(STRSTARTS(STR(?s),
↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
↪ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 19 – People with at least three provably distinct credits

Before / after rows: 0 / 7

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a ex:ThreeCreditContributor .
↪ FILTER(STRSTARTS(STR(?s),
↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
↪ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 20 – Writer-directors

Before / after rows: 0 / 10

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a ex:WriterDirector . FILTER(STRSTARTS(STR(?s),
↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
↪ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 21 – Actor-filmmakers

Before / after rows: 0 / 7

```

PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a ex:ActorFilmmaker . FILTER(STRSTARTS(STR(?s),
↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
↪ ?s rdfs:label ?label } } ORDER BY ?s

```

Query 22 – Award-winning filmmakers

Before / after rows: 0 / 65

```

PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a ex:AwardWinningFilmmaker .
↪ FILTER(STRSTARTS(STR(?s),
↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
↪ ?s rdfs:label ?label } } ORDER BY ?s

```

Query 23 – Award-winning action films

Before / after rows: 0 / 10

```

PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a ex:AwardWinningActionFilm .
↪ FILTER(STRSTARTS(STR(?s),
↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
↪ ?s rdfs:label ?label } } ORDER BY ?s

```

Query 24 – Action/drama crossing films

Before / after rows: 0 / 8

```

PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { ?s a ex:GenreCrossingFilm .
↪ FILTER(STRSTARTS(STR(?s),
↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
↪ ?s rdfs:label ?label } } ORDER BY ?s

```

Query 25 – Nolan contributions through property chain

Before / after rows: 0 / 8

```

PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>

```

```
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { res:person-Q25191 ex:contributedTo ?s .
  ↪ FILTER(STRSTARTS(STR(?s),
  ↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
  ↪ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 26 – Nolan films through inverse director

Before / after rows: 0 / 8

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?label WHERE { res:person-Q25191 ex:directed ?s .
  ↪ FILTER(STRSTARTS(STR(?s),
  ↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/")) OPTIONAL {
  ↪ ?s rdfs:label ?label } } ORDER BY ?s
```

Query 27 – Types present only after reasoning

Before / after rows: 0 / 1261

```
PREFIX ex: <https://laminhtrung.github.io/movie-lod-semantic-web/ontology#>
PREFIX dbo: <http://dbpedia.org/ontology/>
PREFIX res: <https://laminhtrung.github.io/movie-lod-semantic-web/resource/>
PREFIX rdfs: <http://www.w3.org/2000/01/rdf-schema#>
PREFIX rdf: <http://www.w3.org/1999/02/22-rdf-syntax-ns#>
SELECT DISTINCT ?s ?type WHERE {
  GRAPH <urn:movie:reasoned> { ?s a ?type }
  FILTER(STRSTARTS(STR(?s),
    ↪ "https://laminhtrung.github.io/movie-lod-semantic-web/resource/") &&
    ↪ STRSTARTS(STR(?type),
    ↪ "https://laminhtrung.github.io/movie-lod-semantic-web/ontology#"))
  FILTER NOT EXISTS { GRAPH <urn:movie:asserted> { ?s a ?type } }
} ORDER BY ?type ?s
```